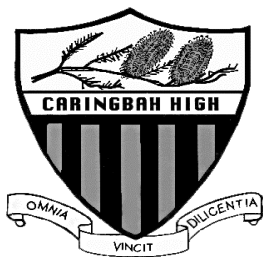


Caringbah High School



2016 **Year 12 Trial HSC** **Examination**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet with standard formulae and integrals is provided.
- In Questions 11–16, show relevant mathematical reasoning and/or calculations.

Total marks – 100

Section I Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–13

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. Given that $z = 5 + 2i$ and $w = i - 3$, what is the value of $2\bar{w} - z$?

(A) $-11 - 4i$ (B) -11
(C) $-1 - 4i$ (D) -1

2. The equation $x^2 - y^2 - 5xy + 5 = 0$ defines y implicitly as a function of x .

What is the value of $\frac{dy}{dx}$ at the point $(1,1)$?

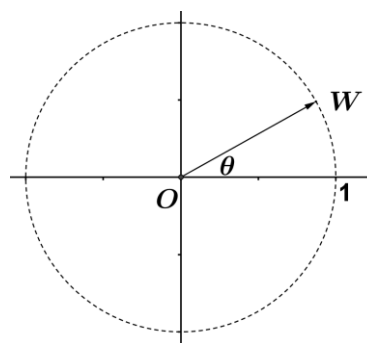
(A) -1 (B) $-\frac{3}{7}$
(C) $\frac{3}{7}$ (D) $\frac{7}{3}$

3. What is the eccentricity of the conic described by the equation $9x^2 - 4y^2 = 1$?

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{5}{3}}$
(C) $\frac{\sqrt{13}}{2}$ (D) $\frac{\sqrt{13}}{3}$

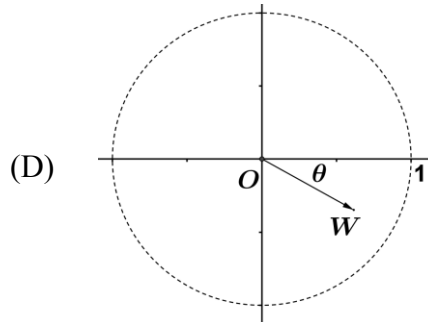
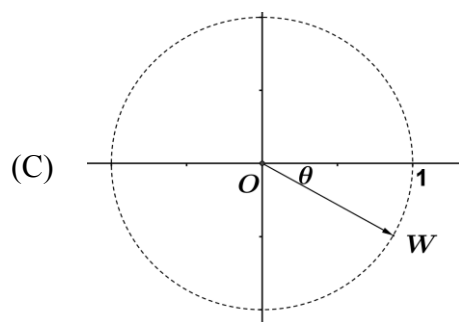
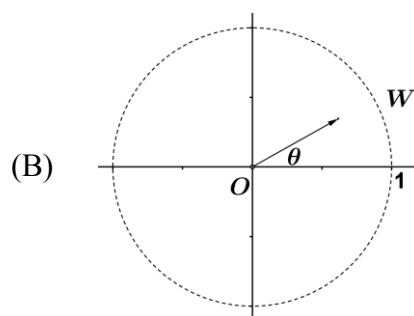
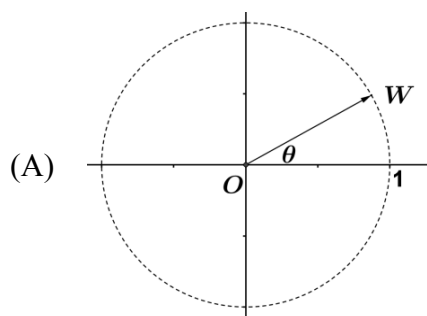
4. $Z = x + iy$ is a complex number, represented on the Argand diagram as shown.

$$|Z| = 1.$$



Which of the following diagrams would represent

the complex number $W = \frac{1}{\bar{Z}}$?



5. The equation $4x^3 - 4x^2 - 15x + 18 = 0$ has a double root at $x = \alpha$.

The value of α is

(A) $\alpha = \frac{-3}{2}$

(B) $\alpha = \frac{3}{2}$

(C) $\alpha = \frac{5}{6}$

(D) $\alpha = \frac{-5}{6}$

6. The conic described by the equation $\frac{x^2}{169} + \frac{y^2}{25} = 1$ has directrices

(A) $x = \pm 12$ (B) $x = \pm \frac{169}{12}$

(C) $y = \pm 12$ (D) $y = \pm \frac{169}{12}$

7. If $z = 2 - \sqrt{12}i$, find the argument of z^5 .

(A) $\frac{-\pi}{3}$ (B) $\frac{-5\pi}{6}$

(C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{6}$

8. The complex number z satisfies $\arg\left(\frac{z-2}{z+2i}\right) = \frac{-\pi}{2}$.

Find the maximum value of $|z|$.

(A) $\sqrt{2}$ (B) $2\sqrt{2}$

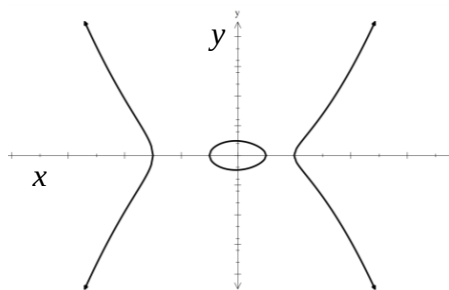
(C) $2 - \sqrt{2}$ (D) $2 + \sqrt{2}$

9. Which of the following is an equivalent expression for $\int \frac{dx}{\sqrt{5-4x-x^2}}$?

(A) $\sin^{-1}(x+2) + C$ (B) $\sin^{-1}\left(\frac{x+2}{3}\right) + C$

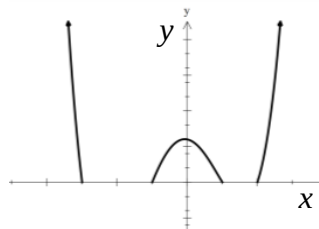
(C) $\sin^{-1}(x-2) + C$ (D) $\sin^{-1}\left(\frac{x-2}{3}\right) + C$

10. $P(x)$ is a polynomial. The graph of $y^2 = P(x)$ is shown below.

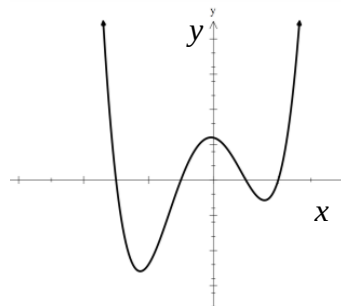


Which of the following graphs is the best representation of $y = P(x)$?

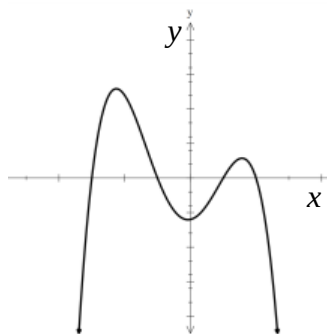
(A)



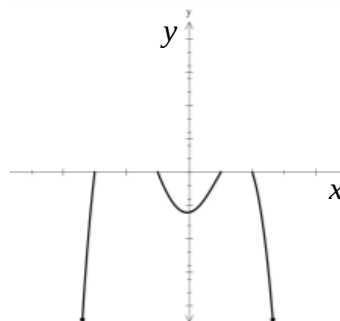
(B)



(C)



(D)



Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Marks
a) If $z = 1 - i$, find z^{-6} in the form $x + iy$.	2
b) If $z = x + iy$, shade on the Argand diagram the region defined by $z\bar{z} \leq 4$.	2
c) Evaluate $\int_{-1}^1 \frac{dx}{x^2 + 2x + 5}$	3
d) Show that $\int_e^{e^4} \ln x \, dx = 3e^4$	3
e) i) Show that $(z - 2i)$ is a factor of $P(z) = z^4 + z^3 + z^2 + 4z - 12$.	1
ii) Hence, find all zeros of $P(z)$.	1
f) Let $A = 3 + 4i$ and $B = 9 + 4i$ be two points on the Argand diagram.	
i) Sketch the locus defined by $ z - A = 5$	1
ii) Draw a clear sketch of the curve defined by $ z - A + z - B = 12$	2

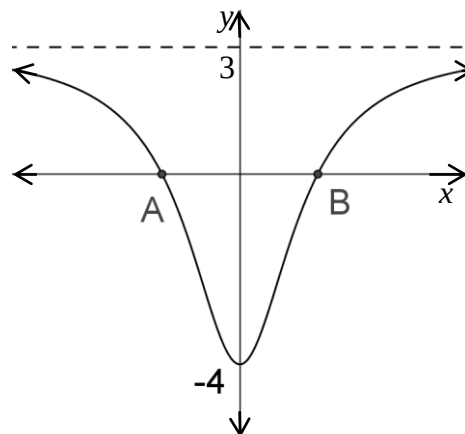
End of Question 11

Question 12 (15 Marks)**Marks**

- a) i) Express $3\sin\theta + 4\cos\theta$ in the form $r\sin(\theta + \alpha)$, where r and α are constants and α is in radians correct to 3 decimal places. 2
- ii) Hence, or otherwise, show that a particle whose displacement x metres, after t seconds, given by $x = 8\cos^2 t + 6\sin t \cos t - 4$ is moving in simple harmonic motion. 2

You may assume that $2\cos A \sin B = \sin(A + B) - \sin(A - B)$

- b) The graph of $y = f(x)$ is shown.



On separate diagrams, show the following graphs, clearly indicating important features.

- i) $y = \frac{1}{f(x)}$ 2
- ii) $y = [f(x)]^2$ 2
- iii) $y = \log_e(f(x))$ 2

Question 12 continues on page 9

Question 12 (continued)

- c) The complex number $z = x + iy$, with x and y real, satisfies $|z - i| = \operatorname{Im}(z)$.
- i) Show that the locus of the point P , representing z , has the 2
Cartesian equation $y = \frac{x^2 + 1}{2}$
- ii) By finding the gradients of the tangents to this curve that pass through 3
the origin, state the set of possible values for the principal argument of z .

End of Question 12

Question 13 (15 Marks)**Marks**

- a) The base of a solid is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. 4

Cross-sections perpendicular to the x -axis are right isosceles triangles with one of the equal sides in the base of the solid.

Show that the volume of the solid is $\frac{8ab^2}{3}$ units³.

- b) Sketch the curve $y = x \ln(x)$, showing any turning points. 2

- c) i) If z is a complex number defined by $z = \cos\theta + i\sin\theta$, 1
show that $\frac{dz}{d\theta} = iz$

- ii) By integrating $\frac{dz}{d\theta} = iz$ with respect to z and θ , 2
show that z can be written in the form $z = e^{i\theta}$.

- d) i) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$ using the substitution $t = \tan \frac{x}{2}$. 3

- ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \cos x + \sin x}$ 3
using the substitution $u = \frac{\pi}{2} - x$.

End of Question 13

Question 14 (15 Marks)

Marks

a) Sketch on the Argand diagram, the locus defined by $\arg(z - 1 + i) = \frac{\pi}{4}$ **1**

b) Two points, $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, lie on the rectangular hyperbola $xy = c^2$.

i) Show that the equation of the tangent at P is $x + p^2y = 2cp$. **2**

ii) The tangents at P and Q meet in T . Find the coordinates of T in terms of c , p and q . **2**

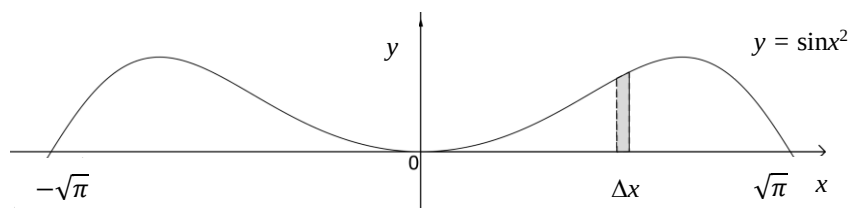
iii) The point T lies on another hyperbola $xy = k^2$, for all positions of P and Q . **1**

Show that $\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$

c) i) Let $I_n = \int_1^e x(\ln x)^n dx$. Show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$, $n = 1, 2, 3, \dots$ **3**

ii) Hence evaluate $\int_1^e x(\ln x)^2 dx$ **2**

d) The area bounded by the curve $y = \sin x^2$ and the x -axis, in the domain $-\sqrt{\pi} \leq x \leq \sqrt{\pi}$, is rotated about the y -axis. **4**



By using the method of cylindrical shells, find the volume of the solid formed.

End of Question 14

Question 15 (15 Marks)**Marks**

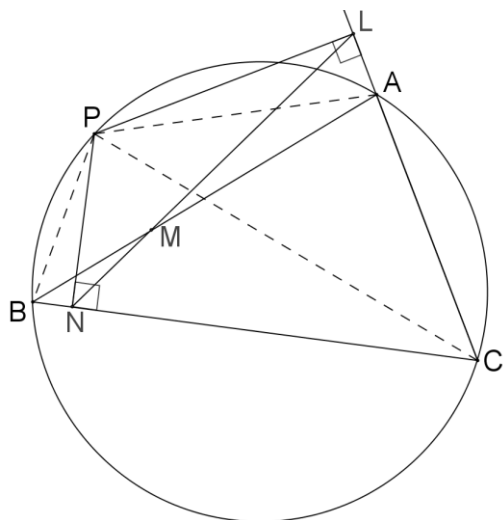
- a) α, β and γ are roots of the cubic equation $x^3 + mx + n = 0$.
- i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ in terms of m and n . 2
- ii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$ in terms of m and n . 2
- iii) Determine the cubic equation with roots α^2, β^2 and γ^2 . 2
- b) For the ellipse $E : 4x^2 + 9y^2 = 36$.
- i) Sketch the ellipse E indicating the position of its foci S and S' , and draw in its directrices. 3
- ii) Show that the point $P(3\cos\theta, 2\sin\theta)$ lies on E . 1
- iii) Derive the equation of the tangent to E at the point P . 2
- iv) Find the coordinates of the point Q where the tangent cuts the major axis. 1
- v) The equation of the normal at P is $\frac{3x}{\cos\theta} - \frac{2y}{\sin\theta} = 5$ and it cuts the major axis at R . A line parallel to the y -axis through P cuts the x -axis at T . Show that $OQ \times RT$ is constant for all positions of P . 2

End of Question 15

Question 16 (15 Marks)

Marks

a)



ABC is an acute-angled triangle inscribed in a circle. P is a point on the minor arc AB of the circle. PL and PN are the perpendiculars from P to CA produce and CB respectively. LN cuts AB at M .

i) Copy the diagram in to your answer booklet. ($\frac{1}{3}$ page)

ii) Explain why $PNCL$ is a cyclic quadrilateral.

1

iii) Hence show $PBNM$ is also a cyclic quadrilateral.

3

iv) Hence show that PM is perpendicular to AB .

2

b) i) Write expressions for $\cos \frac{2\pi}{n}$ and $\sin \frac{2\pi}{n}$ in terms of $\cos \frac{\pi}{n}$ and $\sin \frac{\pi}{n}$

2

ii) By using De Moivre's theorem, show that

3

$$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \left(\cos \frac{\pi}{n}\right)^n$$

c) i) If a and b are both positive numbers, prove that $a + b \geq 2\sqrt{ab}$.

1

Hence, or otherwise, prove that when a , b and c are all positive numbers and $a + b + c = 1$, then

3

$$\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8$$

End of Examination

1) A 2) B 3) C 4) A 5) B

6) B 7) C 8) B 9) B 10) B

Question 11

a) $z = 1 - i = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$

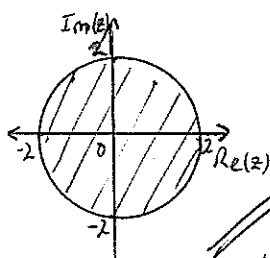
$z^{-6} = (\sqrt{2})^{-6} \operatorname{cis}\left(\frac{6\pi}{4}\right)$

$= \frac{1}{8} \operatorname{cis}\left(-\frac{\pi}{2}\right)$

$= -\frac{i}{8}$

b) $(x+iy)(x-iy) = x^2 + y^2$

$z \cdot \bar{z} \leq 4$



c) $\int_{-1}^1 \frac{dx}{x^2+2x+5} = \int_{-1}^1 \frac{dx}{(x+1)^2+4}$

$= \frac{1}{2} \left[\tan^{-1}\left(\frac{x+1}{2}\right) \right]_{-1}^1$

$= \frac{1}{2} \left[\tan^{-1}1 - \tan^{-1}0 \right]$

$= \frac{\pi}{8}$

d) $\int_e^{e^4} \ln x dx$ $u = \ln x$ $dv = dx$

$du = \frac{1}{x} dx$ $v = x$

$= x \ln x \Big|_e^{e^4} - \int_e^{e^4} x \times \frac{1}{x} dx$

$= [e^4 \ln e^4 - e \ln e] - [x]_e^{e^4}$

$= 4e^4 - e - e^4 + e$

$= 3e^4$

e) $P(2i) = (2i)^4 + (2i)^3 + (2i)^2 + 4 \times 2i - 12$

$= 16 - 8i - 4 + 8i - 12$

$= 0$

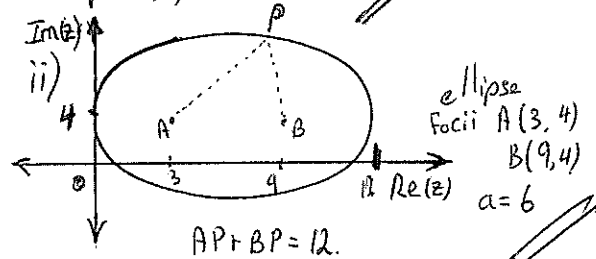
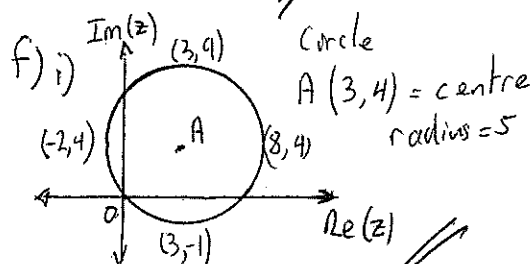
$\therefore (z-2i)$ is a factor of $P(z)$

ii) $P(z)$ has real coefficients $\therefore (z+2i)$ is a factor

$\therefore (z+2i)(z-2i) = (z^2+4)$ is a factor.

$z^4 + z^3 + z^2 + 4z - 12 = (z^2+4)(z^2+z-3)$

roots are $\pm 2i, \frac{-1 \pm \sqrt{1+12}}{2} = \frac{-1 \pm \sqrt{13}}{2}$

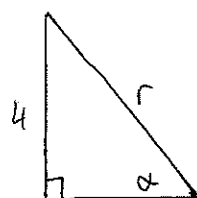


Question 12

a) $3 \sin \theta + 4 \cos \theta$

$= r \sin \theta \cos \alpha + r \cos \theta \sin \alpha$

$r \cos \alpha = 3$ $r \sin \alpha = 4$



$r = 5$ $\tan \alpha = \frac{4}{3}$

$\alpha \approx 0.927$

$3 \sin \theta + 4 \cos \theta = 5 \sin(\theta + 0.927)$

ii) 3

$x = 8 \cos^2 t + 6 \sin t \cos t - 4$

$= 8 \cos^2 t + 3 \sin 2t - 4$

$\dot{x} = 16 \cos t \sin t + 6 \cos 2t$

$= -8 \sin 2t + 6 \cos 2t$

$\ddot{x} = -16 \cos 2t - 12 \sin 2t$

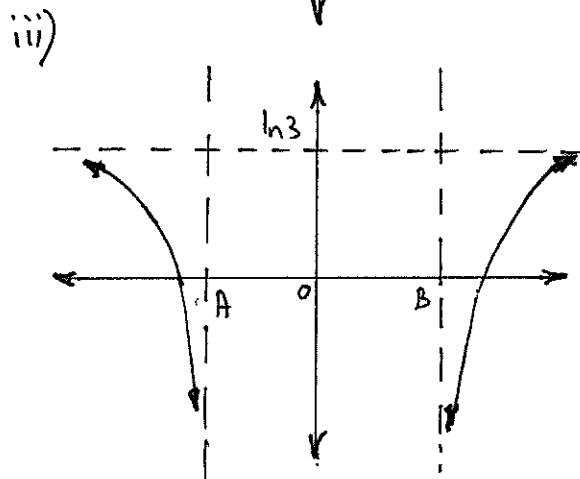
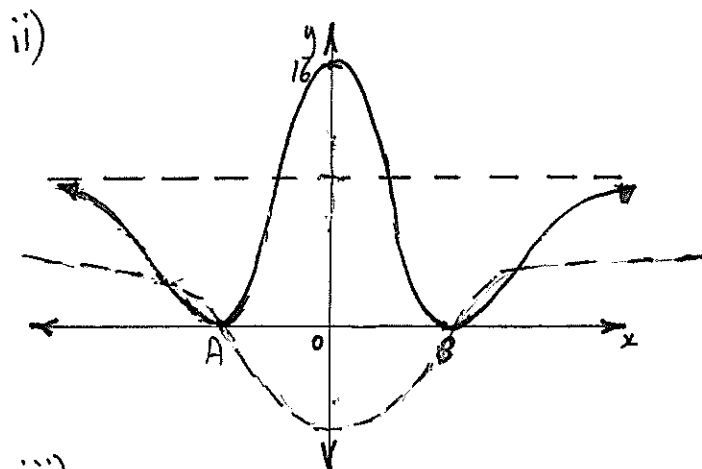
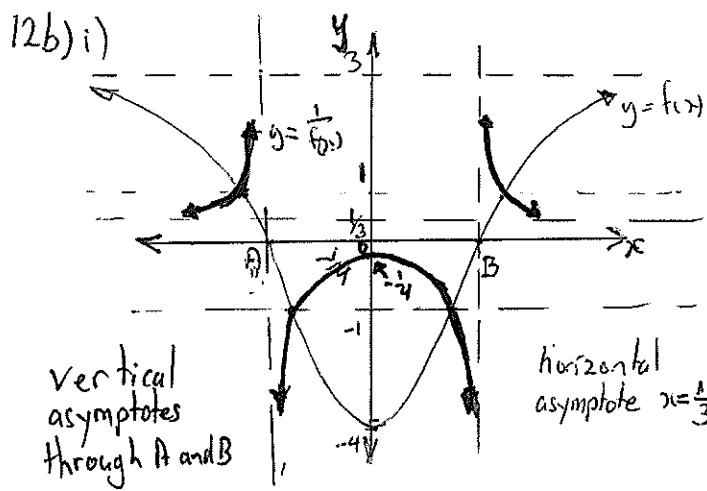
$= -16(2 \cos^2 t - 1) - 24 \sin t \cos t$

$= -32 \cos^2 t - 24 \sin t \cos t + 16$

$= -4(8 \cos^2 t + 6 \sin t \cos t - 4)$

$= -4x$

$\therefore$ SHM, $n=2$, $T = \pi$



c) i) $|z - i| = \text{Im } z$

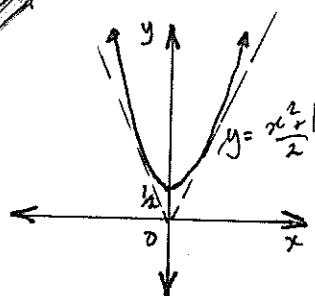
$$\sqrt{(x^2 + (y-1)^2)} = y$$

$$x^2 + y^2 - 2y + 1 = y^2$$

$$x^2 - 2y + 1 = 0$$

$$2y = x^2 + 1$$

$$y = \frac{x^2 + 1}{2}$$



ii) $y = \frac{x^2}{2} + \frac{1}{2}$

$\frac{dy}{dx} = x$ If it passes through 0, then gradient = $\frac{y}{x}$

$$\frac{y}{x} = x$$

$$y = x^2$$

$$\frac{x^2 + 1}{2} = x^2$$

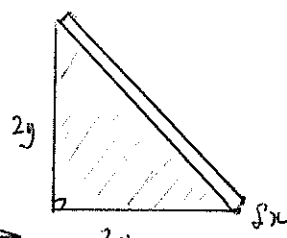
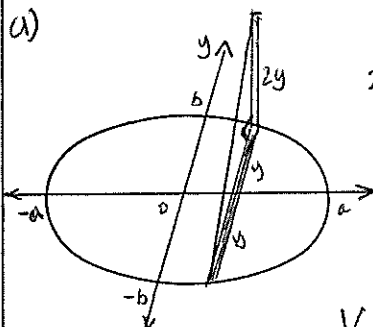
$$x^2 = 1 \therefore x = \pm 1, y = 1$$

ie tangents inclined at $\frac{\pi}{4}$ and $\frac{3\pi}{4}$

$$\therefore \frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$$

Question 13

a)



$$\int V = \frac{1}{2} \times (2y)^2 \int dx$$

$$V = 2 \int_0^a 2y^2 dx$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$$= \frac{b^2}{a^2} (a^2 - x^2)$$

$$V = \frac{4b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$= \frac{4b^2}{a^2} \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \frac{4b^2}{a^2} \left[\left(a^3 - \frac{a^3}{3} \right) - 0 \right]$$

$$= \frac{4b^2}{a^2} \times \frac{2a^3}{3}$$

$$= \frac{8ab^2}{3}$$

b) $y = x \ln x \Rightarrow x > 0$

$$y' = x \times \frac{1}{x} + \ln x$$

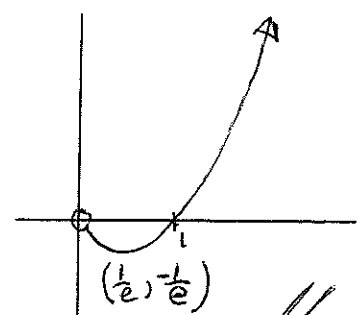
$$= 1 + \ln x$$

$$y' = 0 \ln x = -1$$

$$x = e^{-1}$$

$$= \frac{1}{e}$$

$$y = -\frac{1}{e}$$



$$y'' = \frac{1}{x} \therefore \text{min TP at } \left(\frac{1}{e}, -\frac{1}{e} \right)$$

$$\text{as } x > 0 \Rightarrow y'' > 0$$

ie always concave up.

13 c) i) $z = \cos \theta + i \sin \theta$

$$\frac{dz}{d\theta} = -\sin \theta + i \cos \theta$$

$$i z = i (\cos \theta + i \sin \theta) \\ = i \cos \theta - \sin \theta \\ = \frac{dz}{d\theta}$$

ii) $\frac{dz}{d\theta} = i z$

$$\int \frac{dz}{z} = \int i d\theta$$

$$\ln z = i\theta + c$$

$$\theta = 0 \quad z = \cos 0 + i \sin 0 = 1$$

$$\text{so } \ln 1 = 0 + c \quad \therefore c = 0$$

$$\ln z = i\theta \\ \underline{\underline{z = e^{i\theta}}} \quad \text{[A useful extension to complex numbers!]}$$

d) i)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$$

$$x = \tan^{-1} t \\ \sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$x = 2 \tan^{-1} t$$

$$dx = \frac{2dt}{1+t^2}$$

$$x=0 \quad t=0 \\ x=\frac{\pi}{2} \quad t=1$$

$$\int_0^1 \frac{2dt}{1+t^2 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \times \left(\frac{1+t^2}{1+t^2} \right)$$

$$\int_0^1 \frac{2dt}{1+t^2 + 1-t^2 + 2t}$$

$$\int_0^1 \frac{2dt}{2+2t}$$

$$\int_0^1 \frac{dt}{1+t} = \left[\ln |1+t| \right]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2$$

ii) $\int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \cos x + \sin x} = \int_{\frac{\pi}{2}}^0 \frac{(\frac{\pi}{2} - u) (-du)}{1 + \cos(\frac{\pi}{2} - u) + \sin(\frac{\pi}{2} - u)}$

$$u = \frac{\pi}{2} - x$$

$$du = -dx$$

$$x=0 \quad u = \frac{\pi}{2}$$

$$x = \frac{\pi}{2} \quad u = 0$$

$$= \int_{\frac{\pi}{2}}^0 \frac{-\frac{\pi}{2} du}{1 + \sin u + \cos u} - \int_{\frac{\pi}{2}}^0 \frac{-u du}{1 + \cos u + \sin u}$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{du}{1 + \cos u + \sin u} - \int_0^{\frac{\pi}{2}} \frac{u}{1 + \cos u + \sin u}$$

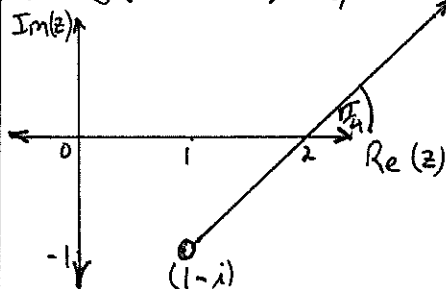
$$\text{let } u = x$$

$$2I = \frac{\pi}{2} \times \ln 2 \quad \text{[from i)]}$$

$$I = \frac{\pi}{4} \ln 2$$

Question 14

a) $\arg(z - (1-i)) = \frac{\pi}{4}$



b) i) $y = \frac{c^2}{x} \quad \frac{dy}{dx} = -\frac{c^2}{x^2}$

$$\text{at } x = cp \quad m = \frac{-c^2}{c^2 p^2} = -\frac{1}{p^2}$$

Eqn of tangent

$$(y - \frac{c}{p}) = -\frac{1}{p^2} (x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp$$

ii) $x + p^2 y = 2cp$ Eqs of tangents

$$x + q^2 y = 2cq$$

$$y(p^2 - q^2) = 2c(p - q)$$

$$y(p - q)(p + q) = 2c(p - q)$$

$$\text{As } (p - q) \neq 0$$

$$y(p + q) = 2c$$

$$y = \frac{2c}{p+q}$$

$$x = 2cp - p^2 y$$

$$= 2cp - \frac{2cp^2}{p+q}$$

$$= \frac{2cp^2 + 2cpq}{p+q} - \frac{2cp^2}{p+q}$$

$$= \frac{2cpq}{p+q} \quad \text{coord of } T \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

iii) $xy = k^2 \Rightarrow \frac{2cpq}{(p+q)} \times \frac{2c}{(p+q)} = k^2$

$$\frac{4c^2 pq}{(p+q)^2} = k^2$$

$$\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$$

14c) i) $I_n = \int_1^e x (\ln x)^n dx$ $u = (\ln x)$ $dv = x dx$
 $du = n(\ln x)^{n-1} \cdot \frac{1}{x} dx$ $v = \frac{x^2}{2}$

$$I_n = \frac{x^2}{2} (\ln x)^n \Big|_1^e - \int_1^e \frac{n x^2}{2} (\ln x)^{n-1} \cdot \frac{1}{x} dx$$

$$= \left[\frac{e^2}{2} (\ln e)^n - \frac{1}{2} (\ln 1)^n \right] - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$\ln e = 1 \quad \ln 1 = 0$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

ii) $n=2$
 $I_2 = \frac{e^2}{2} - I_1$

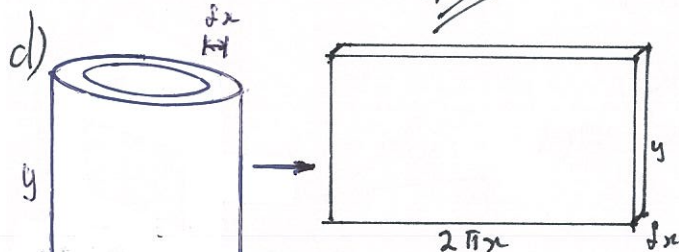
$n=1$
 $I_1 = \frac{e^2}{2} - \frac{1}{2} I_0$

$$I_0 = \int_1^e x (\ln x)^0 dx = \left[\frac{1}{2} x^2 \right]_1^e$$

$$= \frac{e^2}{2} - \frac{1}{2}$$

$$\therefore I_2 = \frac{e^2}{2} - \left[\frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) \right]$$

$$= \frac{e^2}{4} - \frac{1}{4} \text{ or } \frac{e^2-1}{4}$$

d) 

$$r = 2\pi x y \int dx$$

$$V = 2\pi \int_0^{\sqrt{\pi}} x y dx$$

$$= 2\pi \int_0^{\sqrt{\pi}} x \sinh x^2 dx$$

let $u = x^2$ $du = 2x dx$

$x=0$ $u=0$

$x=\sqrt{\pi}$ $u=\pi$

$$V = \pi \int_0^{\pi} \sin u du$$

$$= \pi [-\cos u]_0^{\pi}$$

$$= \pi [1 + 1]$$

$$= 2\pi u^3$$

Question 15

i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma}$

$$x^3 + 0x^2 + mx + n = 0$$

$$-\frac{b}{a} = 0, \quad \frac{c}{a} = m = -\frac{d}{a} = -n$$

$$\text{So } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{m}{n}$$

ii) $\alpha^3 = -m\alpha - n$

$$\beta^3 = -m\beta - n$$

$$\gamma^3 = -m\gamma - n$$

$$\alpha^3 + \beta^3 + \gamma^3 = -m(\alpha + \beta + \gamma) - 3n$$

$$\alpha + \beta + \gamma = 0$$

$$= -3n$$

iii) Sub $x = \sqrt{x}$

$$(\sqrt{x})^3 + m\sqrt{x} + n = 0$$

$$x\sqrt{x} + m\sqrt{x} = -n$$

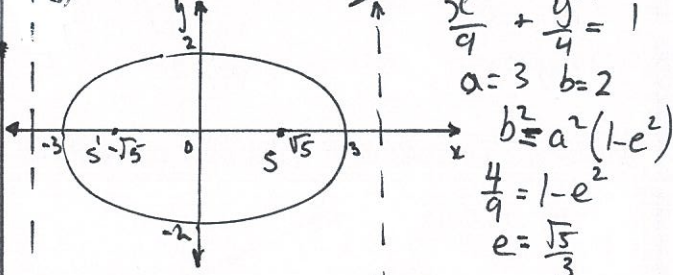
$$\sqrt{x}(x+m) = -n$$

squaring both sides

$$x(x+m)^2 = n^2$$

$$x^3 + 2mx^2 + m^2x - n^2 = 0$$

b)



$$x = -\frac{9}{\sqrt{5}}$$

$$y = \frac{9}{\sqrt{5}}$$

ii) $\frac{(3 \cos \theta)^2}{9} + \frac{(2 \sin \theta)^2}{4} = \frac{9 \cos^2 \theta}{9} + \frac{4 \sin^2 \theta}{4}$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1$$

$\therefore P$ lies on E .

iii) $4x^2 + 9y^2 = 36$

$$8x + 18y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{8x}{18y}$$

$$= -\frac{4x}{9y}$$

at E $m = \frac{-4(3 \cos \theta)}{9(2 \sin \theta)}$

$$m = -\frac{2}{3} \frac{\cos \theta}{\sin \theta}$$

Eqn of tangent

$$(y - 2 \sin \theta) = \frac{-2 \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$$

$$3 \sin \theta y - 6 \sin^2 \theta = -2x \cos \theta + 6 \cos^2 \theta$$

$$2 \cos \theta x + 3 \sin \theta y = 6(\sin^2 \theta + \cos^2 \theta)$$

$$= 6$$

$$\frac{\cos \theta x}{3} + \frac{\sin \theta y}{2} = 1$$

15b i) $y=0$ $2 \cos \theta x = 6$

$$x = \frac{3}{\cos \theta}$$

$$OQ = \frac{3}{\cos \theta}$$

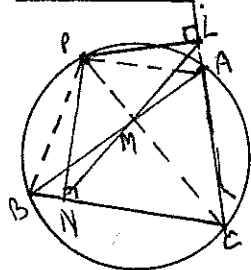
$$OT = 3 \cos \theta$$

$$OR = \frac{5 \cos \theta}{3}$$

$$\begin{aligned} RT &= OT - OR \\ &= 3 \cos \theta - \frac{5 \cos \theta}{3} \\ &= \frac{4 \cos \theta}{3} \end{aligned}$$

$$\begin{aligned} OQ \times RT &= \frac{3}{\cos \theta} \times \frac{4 \cos \theta}{3} \\ &= 4 \text{ ie constant for all positions of } P. \end{aligned}$$

Question 16



ii) $\angle PLC = \angle PNC = 90^\circ$ (perpendicular)
 $\therefore$ opp angles of quadrilateral are supplementary.

$\therefore$ PNCL is a cyclic quad.

iii) As PNCL are concyclic (from ii)

$\angle PNL = \angle PCL$ angles at circumference standing on chord PL

APBC are concyclic so

$\angle PBA = \angle PCA$ angles at circum. standing on chord PA

$\angle PCL$ is also $\angle PCA$

so $\angle PNL = \angle PCL = \angle PCA = \angle PBA$

or $\angle PNL = \angle PBA$

$\angle PNL$ can be named $\angle PNM$

$\angle PBA$ can be named $\angle PBM$

so $\angle PNM = \angle PBM$

$\therefore$ PBNM is a cyclic quad

equal angles at circumference on chord PM.

ii) $\angle BNP$ and $\angle BMP$ stand on chord BP in circle PBNM (from ii).

$\therefore \angle BNP = \angle BMP$

as $PN \perp BC$ (given)

$\angle PMB = 90^\circ$

ie $PM \perp AB$.

b) i) $\cos \frac{2\pi}{n} = \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n}$

$\sin \frac{2\pi}{n} = 2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$

b ii)

$$1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} = \left(\cos^2 \frac{\pi}{n} + i \sin^2 \frac{\pi}{n} \right) + \left(\cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n} \right) + (2i \sin \frac{\pi}{n} \cos \frac{\pi}{n})$$

$$= 2 \cos^2 \frac{\pi}{n} + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}$$

$$= 2 \cos \frac{\pi}{n} \left(\cos \frac{\pi}{n} + i \sin \frac{\pi}{n} \right)$$

$$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \right)^n = 2^n \left(\cos \frac{\pi}{n} \right)^n \left(\cos \frac{\pi}{n} + i \sin \frac{\pi}{n} \right)^n$$

$$= 2^n \left(\cos \frac{\pi}{n} \right)^n \left(\cos \pi + i \sin \pi \right)$$

$$e^{i\pi} = -1$$

$$= -2^n \left(\cos \frac{\pi}{n} \right)^n$$

c) i) $(a-b)^2 \geq 0$

$$(a-b)^2 = a^2 + b^2 - 2ab \geq 0$$

$$(a+b)^2 - 4ab \geq 0$$

$$(a+b)^2 \geq 4ab$$

$$a+b \geq 2\sqrt{ab}$$

(positive case as a, b are both positive)

ii)

$$\left(\frac{1}{a} - 1 \right) = \frac{1-a}{a} \quad \text{If } a+b+c=1$$

$$a = 1-b-c$$

$$= \frac{1-(1-b-c)}{a}$$

$$= \frac{b+c}{a}$$

$$\left(\frac{1}{a} - 1 \right) \left(\frac{1}{b} - 1 \right) \left(\frac{1}{c} - 1 \right) = \frac{(b+c)}{a} \frac{(a+c)}{b} \frac{(a+b)}{c}$$

from i) $a+b \geq 2\sqrt{ab}$

$$\geq \frac{2\sqrt{bc}}{a} \times \frac{2\sqrt{ac}}{b} \times \frac{2\sqrt{ab}}{c}$$

$$\geq \frac{8\sqrt{a^2 b^2 c^2}}{abc}$$

$$\geq \frac{8abc}{abc}$$

$$\geq 8$$